

Ashif Ahmed Shuvo

Machine Learning Enthusiast

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About me

AI & ML expert with a **B.Sc. in Food Engineering**, specializing in **precision and sustainable agriculture and agricultural automation**. Experienced in **computer vision for object detection & recognition**, **IoT-based smart farming solutions**, and **data-driven decision-making for monitoring and optimization**. Passionate about integrating **AI, remote sensing, and sensor-based technologies** to enhance sustainable agriculture and horticulture research.

Skills

Languages: Python, R, Java, C, C++
Computer Vision: EinsteinNet, Vision Transformer, Swin Transformer, DeepLabV3+, ResNet, DenseNet, EfficientNet, SVM, MLP, Feature pyramid Network, U-net, FCN, Attention Mechanism
Packages: TensorFlow, RDKit, OpenCV, Keras API, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn
Data Analysis: Cluster analysis, Regression, PCA, PLS, Multivariate Analysis
Cloud Computing: Microsoft Azure-ML, AWS
Spectroscopy: NIR, Spectrophotometer
Design Tools: Blender

Education

Bangladesh Agricultural University, Mymensingh. M.Sc. in Agrometeorology	March 2025 - Present
Bangladesh Agricultural University, Mymensingh. B.Sc. in Food Engineering	Jan 2018 - Oct 2024
Government M. M. College, Jessore. Higher Secondary Certificate (HSC)	April 2014 - April 2016
Jessore Zilla School, Jessore. Secondary School Certificate (SSC)	Jan 2006 - Feb 2014

Publications & Writings

- **Shuvo, Ashif Ahmed**, Bhuian, Wahada Jinnat Oishy, Rahman, Afzal, Iqbal, Abdullah, *EinsteinNet and State-of-the-Art ML Models for Android-Based Orange Classification: Integration, Evaluation, and Deployment* (2025). **Smart Agricultural Technology**. <https://doi.org/10.1016/j.atech.2025.101072>
- **Shuvo, Ashif Ahmed**, *Comparative Analysis of Regression Models for Commodity Price Prediction* (2025). SSRN. <https://doi.org/10.2139/ssrn.5248271>.
- **Shuvo, Ashif Ahmed**, Bhuian, Wahada Jinnat Oishy, Rahman, Afzal and Iqbal, Abdullah. *Integration of Machine Learning in an Android Application for Automated Orange Identification and Classification* (2025). SSRN. <https://doi.org/10.2139/ssrn.5183017>.
- **Shuvo, Ashif Ahmed**, Monika, Subrina Afrin, Turza, Mahadi Abser, Iqbal, Md. Asif, Begum, Ismat Ara, Alam, Mohammad Jahangir. *Forecasting Wheat Flour Prices in Bangladesh to 2030: A Bayesian-Optimized XGBoost Framework with Structural Break Detection, Ensemble Methods, and Comprehensive Robustness Validation* (2026). (Manuscript under preparation).
- **Shuvo, Ashif Ahmed**, Hasan, Sakibul, Anjum, Abir, Shahed, Asaduzzaman, Al-Mamun, Mohammad, Al-Mamun, Mohammad. *Deep Learning-Based Weight Prediction for Indigenous Pabna Cattle: A Comparative Analysis of Seven CNN Architectures with TensorFlow.js Deployment*

and Metabolically-Scaled Phytogetic Supplement Integration (Smart-Herbo) (2026). (Manuscript under preparation).

Projects

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- CatalystLab** (<https://catalystlab.tech>) Mar 2026 - Present
- **Full-stack AI research SaaS:** Next.js 14, Firebase, Gemini 2.5 Flash API; 20 AI instruments, 9 academic API integrations, 250M+ papers, and a free model.
- Smart-Herbo** (<https://github.com/myself-aas/Smart-Herbo>) June 2025 - Feb 2026
- Built a privacy-first, on-device livestock management application leveraging TensorFlow.js for cattle weight prediction and metabolic scaling-based Plantain Herb nutritional recommendations, enabling precision agriculture without cloud dependency.
- Wine Quality Prediction** (<https://github.com/myself-aas/Wine-Quality-Prediction>) Oct 2024 - Dec 2024
- Examined how physicochemical properties influence wine quality, offering actionable insights for the wine industry to optimize product quality.
- Study Duration Prediction** (<https://github.com/myself-aas/Study-Duration-Prediction>) Nov 2024
- Developed a machine learning-powered web application using Flask and Python to predict study duration based on factors like GPA, social media engagement, and lifestyle influences, providing data-driven insights into academic performance determinants.
- International Nutrition Olympiad, Sher-e-Bangla Agricultural University** Sep 2022 - Oct 2022
- Developed a mobile application for BMI calculation & healthy weight plan.

Experiences

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- Research Assistant**, Bangladesh Agricultural University Research System Mar 2025 - Present
- Developed ML models for division-wise agricultural commodity price forecasting using spatio-temporal data in Bangladesh.
- Internship**, Data Analytics and Visualization Job Simulation, Accenture Nov 2024
- Successfully completed practical tasks in Project Understanding, Data Cleaning & Modeling, Data Visualization & Storytelling, Present to the Client ([Certificate of Completion](#)).
- Diploma**, Python Development Essentials, Institute of MTF Oct 2024 - Nov 2024
- Gained proficiency in data structures, functions, and OOP ([Certificate of Completion](#)).
- Internship**, Data Analytics, Quantum Sep 2024 - Nov 2024
- Analyzed transaction data to identify benchmark stores and deliver insights for strategic commercial decisions ([Certificate of Completion](#)).
- Internship**, Akij Food and Beverages Ltd., Bangladesh. June 2024 - July 2024
- Completed a 15-day internship at AFBL focused on mango pulp, chutney, and mustard oil production, emphasizing HACCP, quality control, and logistics.

Achievements

Bangladesh Government Merit Scholarship at JSC, SSC, & HSC exams.

Professional Development & Online Learning

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1. **OpenCV Bootcamp** from [OpenCV University](#) (Grade Earned - 100%).
 2. **TensorFlow Keras Bootcamp** from [OpenCV University](#) (Grade Earned - 100%).
 3. **AI Agents Fundamentals** from [Hugging Face](#).
 4. **Intro to Programming** from [Kaggle](#).
 5. **Python** from [Kaggle](#).
 6. **Machine Learning Explainability** from [Kaggle](#).
 7. **Machine Learning Online Course** from MakeIntern through [Udemy](#).

8. **Data Science Career Path** from MakeIntern through [Udemy](#).
9. **Internet of Things (IoT) Online Course** from MakeIntern through [Udemy](#).
10. **Python for OOP: The A-to-Z OOP Python Programming Course** from Meta Brains through [Udemy](#).
11. **Python & TensorFlow: Deep Dive into Machine Learning** from Meta Brains through [Udemy](#).
12. **Python for Deep Learning: Build Neural Networks in Python** from Meta Brains through [Udemy](#).
13. **Harnessing AI for a social impact** from [Udemy](#).
14. **Harnessing AI and Machine Learning for Geospatial Analysis** from [Udemy](#).

Leadership & Extracurricular Activities

Vice President , Nutrition Club, BAU	Apr 2023 - Present
• Lead 2 national-level competition • Arranged more than 20 workshops and seminars	
IT Advisor , Shwapno Youth Development Organization.	July 2020 - Nov 2021
• Provided emergency support during COVID-19 pandemic • Organized around 15 online training sessions	
Campus Ambassador , English Olympiad	Jan 2019 - Dec 2020
• Co-ordinated teams from 40 educational institutions • Arranging competition	
General Member , Volunteer for Bangladesh.	Feb 2019 - Oct 2020
• Fundraising for unprivileged children • Several social activities and awareness problems	

Academic Reference

Dr. Ahmed Khairul Hasan

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Dr. Mohammad Jahangir Alam

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